

Fall 2025 Midterm I

The exam is closed book and closed notes. You are allotted one 8.5" x 11", single sided, "cheat sheet" of handwritten notes. You may also use a simple scientific (non-graphing) calculator. Your phone is not allowed as a calculator. Using any resources outside of the aforementioned items is strictly prohibited.

While you take the exam, you are prohibited from discussing the test with anyone. If you are taking the test after your classmates, you are also prohibited from talking to them about the test before you take it. Evidence of cheating may result in a 0 on the exam and be reported to the Student Conduct Board.

First and Last Name (print clearly): _____

Student ID Number (print clearly): _____

Please read the following statement and sign below to indicate that you understand the policies for this exam
You must show your student ID when you submit your test.

UC Berkeley Honor Code

"As a member of the UC Berkeley community, I act with honesty, integrity, and respect for others." Please carefully read the statements below, and indicate your understanding and intent to adhere to the UC Berkeley Honor code by signing your name in the space below. I agree not to engage in any of the following behaviors:

- Copying or attempting to copy from others during an exam or on an assignment.
- Communicating answers with another person during an exam.
- Pre-programming a calculator or other personal electronic device to contain answers, or using other unauthorized information for exams.
- Using unauthorized materials, i.e. prepared answers.
- Allowing others, or utilizing software like generative AI, to do an assignment or a portion of an assignment for you, including the use of a commercial term-paper service.
- Submitting the same assignment for more than one course without prior approval of all the instructors involved.
- Collaborating on an exam or assignment with any other person without prior approval from an instructor.
- Taking an exam for another person or having someone take an exam for you.
- Altering a previously graded exam or assignment for the purpose of a grade appeal or of gaining points in a re-grading process.
- Submitting an electronic file the student knows to be unreadable or corrupted instead of a completed assignment.

Signature:

Sign your name:

INSTRUCTIONS:

Handwrite your responses using a pencil or pen in the space provided. Give your responses **ONLY** in the space provided. Do not write your responses outside of the provided spaces. The additional space provided, including space on the back side of the exam can be used as scratch paper but will not be graded.

Phones should be turned OFF prior to the start of the exam and secured in your backpack or another secure location. Do not leave your phone or other electronic devices out. If you need to leave the room for any reason during the exam please flag a GSI and let them know prior to exiting the room. Upon leaving the exam room, you must leave your electronic device(s) with the GSIs. Time will still accrue when you leave the room.

The length of the midterm is 50 minutes. If you finish early and are satisfied with your work you may leave early. Hand in your exam to a GSI, who will verify that they received it, with your student ID.

- Unless otherwise specified in the question, format your answers according to the following guidelines:
 - present your answers rounded to two decimal places
 - present proportions as decimal values rounded to 2 decimal places rather than percents (ex: 0.41 rather than 41.30%)
- All logarithms (“logs”) are the natural logarithm (base e ; $\log_e(x)$; $\ln(x)$)

Exam Format:

Short Format Questions: Q1-5 [5 points]

Long Format Question: Q6 [7 points]

Long Format Question: Q7 [4 points]

Long Format Question: Q8 [6 points]

Long Format Question: Q9 [8 points]

Extra credit: Q10 [1 point]

Optional Feedback Question

Short Format Questions: Please select only one answer choice.

1. [1 point] True or False: The PPDAC statistics method stands for Purpose, Plan, Design, Analysis, Conclusion.

- ☐ A. True
- ☐ B. False

2. [1 point] The Bears are on a roll! Cal Men's football is having a great start to their season, so you'd like to visualize points scored per game over the course of the season. Which ggplot function is most appropriate?

- ☐ A. `geom_point()`
- ☐ B. `geom_bar()`
- ☐ C. `geom_histogram()`
- ☐ D. `geom_line()`

3. [1 point] Which statement about Pearson's correlation coefficient r is NOT correct?

- ☐ A. r always takes values between -1 and 1
- ☐ B. r measures the strength and direction of a linear relationship between two quantitative variables
- ☐ C. r is resistant to outliers
- ☐ D. r best describes linear relationships only
- ☐ E. A strong correlation does not imply that one variable causes the other

4. [1 point] Which of the following best illustrates Simpson's Paradox?

- ☐ A. The correlation between hours studied and exam score is positive in both men and women, but slightly weaker in men.
- ☐ B. A drug appears to reduce mortality within both younger and older patient groups, but when the groups are combined, the overall mortality is higher in the treatment group.
- ☐ C. A study finds that students with higher GPAs also tend to have higher SAT scores, but the relationship is not perfect.
- ☐ D. Taller individuals tend to weigh more on average, regardless of sex.

5. [1 point] You hypothesize that physical activity improves self-esteem. To test this, you conduct a randomized experiment: one group receives a physical activity intervention, while another group has their activity level controlled. You then measure changes in self-esteem. What type of problem is this study?

- ☐ A. Descriptive
- ☐ B. Causative
- ☐ C. Predictive

Long Format Questions

Q6 [7 points]

Obesity is defined as having a Body Mass Index (BMI) ≥ 30 kg/m². According to the CDC, adult obesity is associated with an increased risk of chronic diseases such as type 2 diabetes. A study investigates the association between obesity and type 2 diabetes among 2,400 U.S. adults. Diabetes is defined according to ADA diagnostic criteria. Obesity status was determined by measured height and weight at the baseline visit.

Source: Adapted from CDC National Diabetes Statistics Report.

Use the following facts to complete the table below:

- 25% of the adults are obese
- 12% of the adults have diabetes
- Among the non-obese adults, 8% have diabetes

	Diabetes	No Diabetes	
Obese	A	???	B
Not Obese	C	???	???
	D	???	2400

6.1 [4 points] Fill in correct values

- A: _____
- B: _____
- C: _____
- D: _____

6.2 [2 points] Name one potential confounder if the study were causative. Justify in 1-2 sentences.

6.3 [1 point] In no more than one sentence, describe a potential issue with the obesity measure:

Q7 [4 points]

Read the adapted abstract below and answer the following questions.

Background: Exposure to blue light from phones and laptops late at night has been linked to poor sleep quality and impaired concentration. We aimed to assess the effects of different blue-light reduction strategies, individually or in combination, on sleep and academic performance among college students.

Methods: In this cluster-randomized controlled trial, we enrolled first-year student dormitories at three universities in California, with an average of 25 students per dorm. Groups of dorms were block-randomized into four intervention groups and one control group. The four intervention groups were: use of blue-light-filtering apps on devices; wearing blue-light-blocking glasses; a digital curfew program encouraging students to stop device use one hour before bedtime; and a combined intervention of filters, glasses, and curfew. The control group received no intervention. Outcomes were measured at baseline and after 3 months: average nightly sleep duration and quality (measured by actigraphy and surveys) and GPA at the end of the semester. Masking of participants was not possible.

7.1 [1 point] What type of a problem is this study interested in addressing? (Options: Descriptive, Predictive, Causative/Etiologic)

7.2 [1 point] This is a(n) _____ study design. (Options: Observational, Experimental, Case-Control, Cross-Sectional)

7.3 [1 point] What type of variable is the exposure of interest? (Options: Ordinal, Nominal, Continuous, Discrete)

7.4 [1 point] Who is the target population of the study?

Q8 [6 points]

The Tang Center at UC Berkeley conducted a small study on student health. The dataset `berkeley_data` summarizes results across different variables. Each record contains the student's major, sex, race, self-reported well-being score (0–1).

```
head(berkeley_data)
```

```
## # A tibble: 6 x 4
##   major      sex    race  well_being
##   <chr>    <chr> <chr>    <dbl>
## 1 Biology   Female White     0.76
## 2 Public Policy Female White     0.68
## 3 Public Health Male    Latino     0.8
## 4 Biology   Female Asian     0.7
## 5 Engineering Female Latino    0.79
## 6 Biology   Female Black     0.78
```

8.1 [2 points] We want to view well-being as a percentage, rather than a proportion. Fill in the code below that will create a new column called `well_being_100` that will perform this transformation and assign it to a new dataframe called `berkeley_data_new`.

```
berkeley_data_new <- -----
```

Researchers at the Tang Center wrote the following code chunk to better understand the distribution of `well_being_100`:

```
summary <- berkeley_data_new %>%
  summarize(
    min = min(well_being_100),
    Q1 = quantile(well_being_100, 0.25),
    median = median(well_being_100),
    Q3 = quantile(well_being_100, 0.75),
    max = max(well_being_100),
    IQR = IQR(well_being_100))
```

```
summary
```

```
## # A tibble: 1 x 6
##   min    Q1 median    Q3    max    IQR
##   <dbl> <dbl>  <dbl> <dbl> <dbl> <dbl>
## 1    60    69    72    75    95     6
```

8.2 [2 points] The Tang Center has asked you to draw a box plot by hand. You have all of the necessary information to create this plot except for the upper and lower whisker values. Calculate them below (Show your work!)

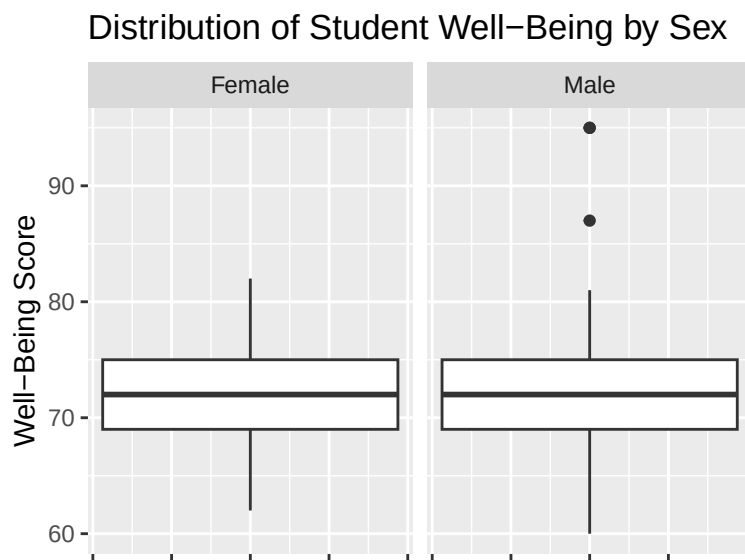
Lower whisker:

Upper whisker:

8.3 [1 point] Values that fall below the lower whisker and above the upper whisker are called _____?

Lastly, researchers wanted to visualize the distribution of well-being stratified by sex:

```
well_being_plot_sex
```



8.4 [1 point] Fill in the blank below with code that stratifies the plots by sex.

```
well_being_plot_sex <- ggplot(berkeley_data_new, aes(y = well_being_100)) + geom_boxplot()
+ _____
```

Q9 [8 points]

Read the adapted abstract below and answer the following questions.

Abstract: Depression is a leading cause of disability worldwide. Researchers examined whether physical activity is associated with lower depressive symptoms. In a cross-sectional analysis, physical activity levels were inversely correlated with depression symptom scores ($r \approx -0.16$, $p < .001$), suggesting that more activity is associated with fewer depressive symptoms.

*Abstract adapted from Pascoe et al., 2024 (“Physical activity and depression symptoms in people with osteoarthritis-related pain: A cross-sectional study”).

The output below shows a preview of the researchers’ dataset `pa_dep_dat`.

```
head(pa_dep_dat)
```

```
## # A tibble: 6 x 3
##   id phys_act_hours dep_score
##   <int>      <int>    <dbl>
## 1     1          0     17.4
## 2     2          0     17.8
## 3     3          1     18.8
## 4     4          1     17.3
## 5     5          2     16.5
## 6     6          2     18.1
```

9.1 [3 points] Fill in the two lines of code that will give you the output of the results below

```
## # A tibble: 2 x 5
##   term          estimate std.error statistic  p.value
##   <chr>          <dbl>    <dbl>    <dbl>    <dbl>
## 1 (Intercept)    18.3      0.414     44.1 8.47e-20
## 2 phys_act_hours -0.827    0.0775    -10.7 3.28e- 9
```

```
pa_lm <- __1__(__2__ ~ __3__, __4__ = pa_dep_dat)
__5__(__6__)
```

- 1: _____
- 2: _____
- 3: _____
- 4: _____
- 5: _____
- 6: _____

9.2 [2 points] Interpret the slope in the context of the problem in one sentence.

9.3 [1 point] Interpret the intercept in the context of the problem in one sentence.

9.4 [2 points] Predict the approximate depression score for a student who reports 7 hours of physical activity per week. Show your work, and round your answer to two decimal places.

10.[1 point] Extra Credit: In one sentence, describe a core principle of the Nuremberg Code

Optional feedback

Take a moment to let us know any issues that came up on the exam.

This space is intended for scratch paper, we will NOT be grading this page